

## Homework #6

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## Question 5.

**(a):**  $5n^3 + 2n^2 + 3n = \Theta(n^3)$

For  $T(n) = 5n^3 + 2n^2 + 3n$  and  $h(n) = n^3$ ,  $5n^3 + 2n^2 + 3n = \Theta(n^3)$  if and only if  $T(n) = O(h(n))$  and  $T(n) = \Omega(h(n))$ .

For  $n \geq 1$ ,  $n \leq n^2 \leq n^3$ .

$$\begin{aligned} 5n^3 + 2n^2 + 3n &\leq 5n^3 + 2n^3 + 3n^3 \\ 5n^3 + 2n^2 + 3n &\leq 10n^3 \\ 5n^3 + 2n^2 + 3n &\leq 10h(n) \end{aligned}$$

$T(n) = O(h(n))$  since there are positive constants  $c = 10$  and  $n_0 = 1$  such that  $T(n) \leq ch(n)$  when  $n \geq n_0$ .

For  $n \geq 1$ ,  $2n^2 + 3n \geq 0$ .

$$\begin{aligned} 2n^2 + 3n &\geq 0 \\ 5n^3 + 2n^2 + 3n &\geq 5n^3 \\ 5n^3 + 2n^2 + 3n &\geq 5h(n) \end{aligned}$$

$T(n) = \Omega(h(n))$  since there are positive constants  $c = 5$  and  $n_0 = 1$  such that  $T(n) \geq ch(n)$  when  $n \geq n_0$ .  
 $5n^3 + 2n^2 + 3n = \Theta(n^3)$  since  $T(n) = O(h(n))$  and  $T(n) = \Omega(h(n))$ .

**(b):**  $\sqrt{7n^2 + 2n - 8} = \Theta(n)$

For  $T(n) = \sqrt{7n^2 + 2n - 8}$  and  $h(n) = n$ ,  $\sqrt{7n^2 + 2n - 8} = \Theta(n)$  if and only if  $T(n) = O(h(n))$  and  $T(n) = \Omega(h(n))$ .

For  $n \geq 1$ ,  $n \leq n^2$ .

$$\begin{aligned} \sqrt{7n^2 + 2n - 8} &\leq \sqrt{7n^2 + 2n^2} \\ \sqrt{7n^2 + 2n - 8} &\leq \sqrt{9n^2} \\ \sqrt{7n^2 + 2n - 8} &\leq 3n \\ \sqrt{7n^2 + 2n - 8} &\leq 3h(n) \end{aligned}$$

$T(n) = O(h(n))$  since there are positive constants  $c = 3$  and  $n_0 = 1$  such that  $T(n) \leq ch(n)$  when  $n \geq n_0$ .

For  $n \geq 4$ ,  $2n - 8 \geq 0$ .

$$\begin{aligned} \sqrt{2n - 8} &\geq \sqrt{0}. \\ \sqrt{7n^2 + 2n - 8} &\geq \sqrt{7n^2}. \\ \sqrt{7n^2 + 2n - 8} &\geq \sqrt{7}n. \\ \sqrt{7n^2 + 2n - 8} &\geq \sqrt{7}h(n). \end{aligned}$$

$T(n) = \Omega(h(n))$  since there are positive constants  $c = \sqrt{7}$  and  $n_0 = 4$  such that  $T(n) \geq ch(n)$  when  $n \geq n_0$ .  
 $\sqrt{7n^2 + 2n - 8} = \Theta(n)$  since  $T(n) = O(h(n))$  and  $T(n) = \Omega(h(n))$ .